VTPEH 6270

Class 11

Shiny App

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House Keeping…

By Next Time (Tuesday April 21)

- Progress on Check Point #07 (due that week)
- Need help? Office hours on Wednesdays, 12:30-1:30 PM in S1-122
  - Room booked 12:00-2:00 PM if you want a place to work
  - Can’t make that time? Email Anika or Jinglei

- Next session: Shiny App Peer-Review
  - Review and crash-test apps in groups, using the Check Point Rubric as a guide
  - This Check Point is similar to the Final App
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Check Point #07 Content

- You are free to develop the app you want, as long as it respects those conditions:
  - Focuses on a **question related to data you chose for your project**.
  - Includes a **description of the app goal(s)**.
  - Includes at least one (real or simulated) **data visual**.
  - Includes at least one **button** for the user to interact with, with impact on other elements of the app (e.g., visual or text).
- Suggestions:
  - Data exploration, with the possibility for the user the pick the type or plot.
  - Data analysis, with possibility for the user to pick either the test, or the variable, or a specific subset of the data.
  - Simulations for power analysis, with possibility for the user the explore different parameters.

Check Point #07 Grading

- Check the Rubric!
- Use as a check list to build your app

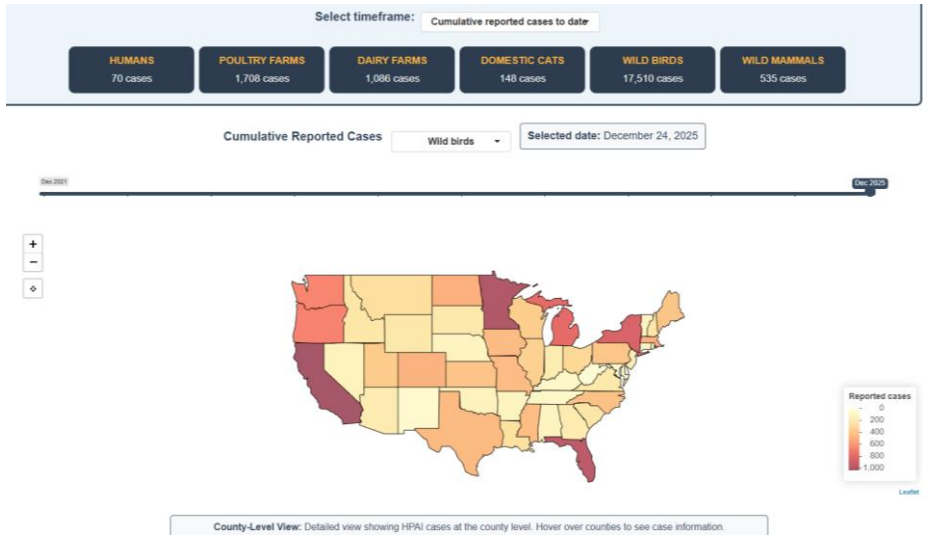
Context & References	Excellent!	Good!	Should be improved	No marks
	The web page includes a short description of the context (e.g., motivating question), an author list, information regarding the data source, information regarding the methods, a link to the GitHub repository, and/or an AI disclosure statement as appropriate. 3 pts	The web page is missing some elements, but broad context is still understandable (e.g., by looking for the information in the linked report). 2 pts	The web page is missing some elements, impacting the understandability of the broad context. 1 pts	The web page misses several elements, and the broad context is unclear. 0 pts

Shiny App

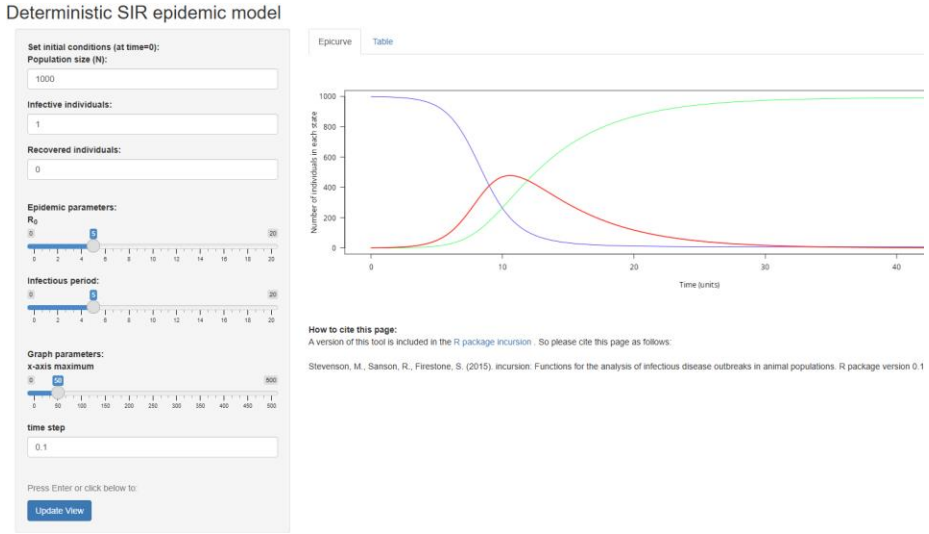
Getting Ready

- Set up a shinyapps.ioLinks to an external site. account
- `install.packages("shiny")`
- `install.packages("bslib")`
- `install.packages("rsconnect")`

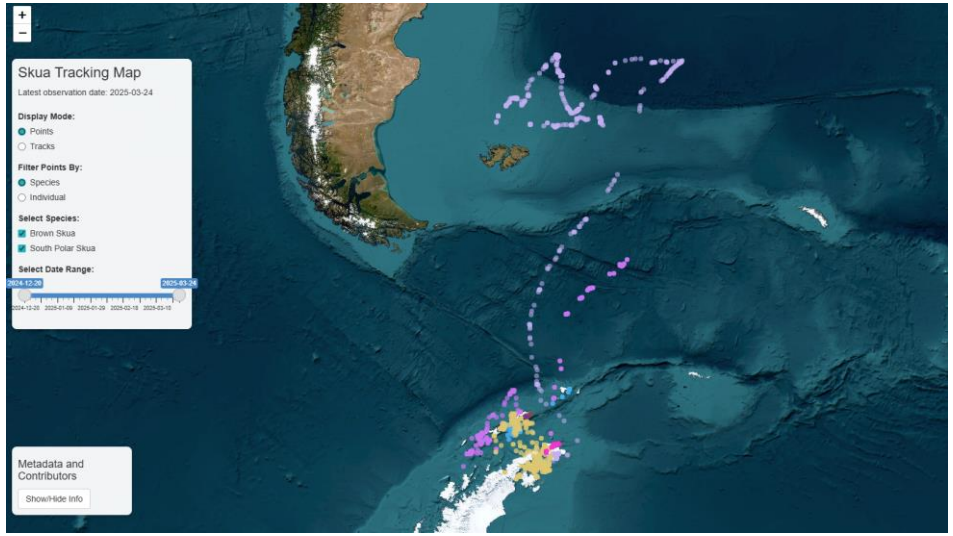
Examples



vet.cornell.edu/highly-pathogenic-avian-influenza-bird-flu-resource-center

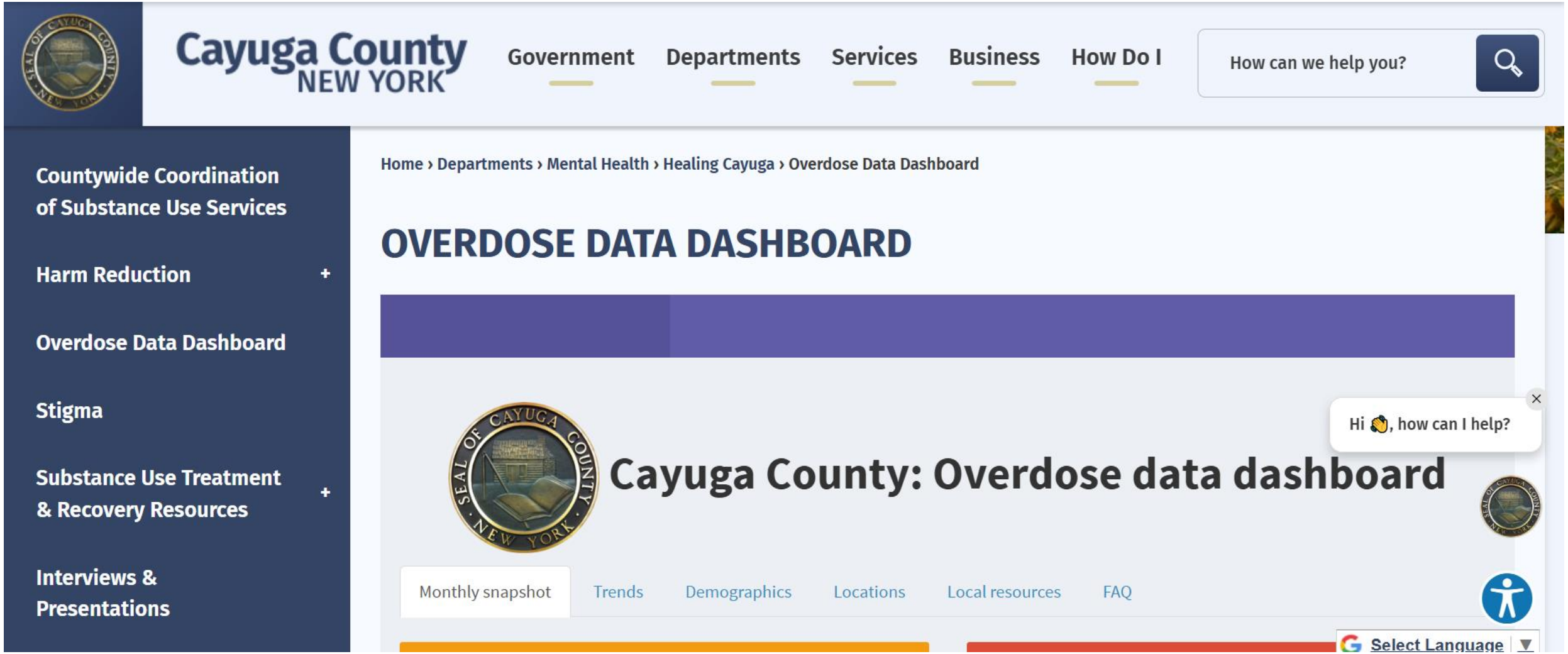


shiny.vet.unimelb.edu.au/epi/sir.vanilla



bcs98.shinyapps.io/antarctic_skua_tracking_app

Examples



cayugacounty.gov/1634/Overdose-Data-Dashboard

Getting Started

- Full tutorial: shiny.posit.co/r/getstarted
- Step-by-step guide: statsandr.com/blog/how-to-publish-shiny-app-example-with-shinyapps-io/#step-by-step-guide

Some Coding Advice

- Work from a repository connected to GitHub
- Do not work from a cloud to avoid connectivity issues
- When deploying, make sure you work from the repository that contains the App.R file

- Make sure all dependencies (other scripts, pictures, data, etc… are in the same folder)

- Start from a Shiny tutorial, and adjust step-by-step to match your project

Code Structure

- ui (user interface): what the users see
  - Purpose:
    - Layout: How elements are arranged on the page
    - Input widgets: Buttons, sliders, text boxes, dropdowns
    - Output placeholders: Where plots, tables, text will appear
    - Styling: Colors, fonts, CSS styling
  - Characteristics:
    - Static structure: Defines the framework/skeleton
    - No computation: Just describes the layout
    - User interaction points: Where users can provide input

Code Structure

- server: what happens in the background
  - Purpose:
    - Reactive logic: Responds to user input changes
    - Data processing: Calculations, filtering, analysis
    - Generate outputs: Creates plots, tables, text
    - Business logic: The "brain" of your app
  - Characteristics:
    - Dynamic: Changes based on user input
    - Reactive: Automatically updates when inputs change
    - Computational: Where all the work happens

- Line of code to generate the app

How to Make a Good App

- The context and goal are clear
- The resources (data, code…) can easily be found
- Use is intuitive and easy
  - See uxdesign.cc/the-worst-volume-control-ui-in-the-world-60713dc86950
- All parameters, links, etc… have been tested to avoid bugs

Questions?

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